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Canadian Bioinformatics Workshops



Module 5: Single Nucleotide Variants

Dr. Ido Hatam

CBW High-Throughput Genomics

November 4-6, 2026

Langara.

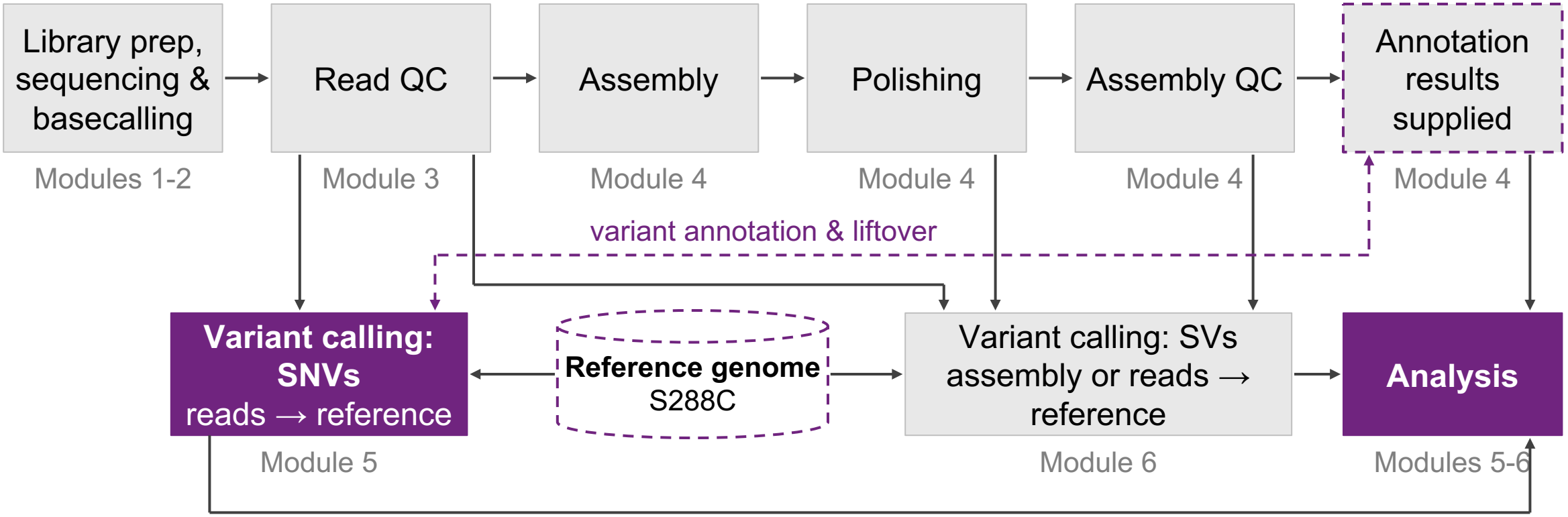
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Learning Objectives

- Explain what SNVs (and small indels) are, their types, and what they are used for
- Explain how SNVs are called from reads aligned to a reference, and why reads are used rather than the assembly
- Read a VCF record (CHROM, POS, REF, ALT, QUAL, FILTER, INFO, FORMAT/GT)
- Explain why short-read variant calling needs extra steps
- Annotate SNVs with snpEff and interpret their predicted impact


The Corgi Yeast
S288C (Reference Genome)
Well annotated
Predictable
Domesticated
A model organism
Plays well with others
REFERENCE GENOME
S288C
LAB FAVOURITE
BREAD BEER WINE SCIENCE
LAB SWEET LAB
SGD
YEAST GENOME
STANDARD PROTOCOLS
REPRODUCIBLE SCIENCE
GOOD YEAST GOOD SCIENCE
Same species. Different life choices.
VS
COMPARATIVE GENOMICS
SAME SPECIES. DIFFERENT WORLDS.
The Wolf Yeast
BAL_1a (Wild Isolate)
Novel alleles
Structural variants
Environmental adaptation
Wild genome
Survived (real) life
NOVEL VARIATION INSIDE
BAL_1a
NATURE SELECTS DIFFERENTLY
TEs
SNPs
CNVs
INDELs
and more...
DOMESTICATED. WILD. SAME YEAST. A BIGGER STORY.
canadian bioinformatics hub

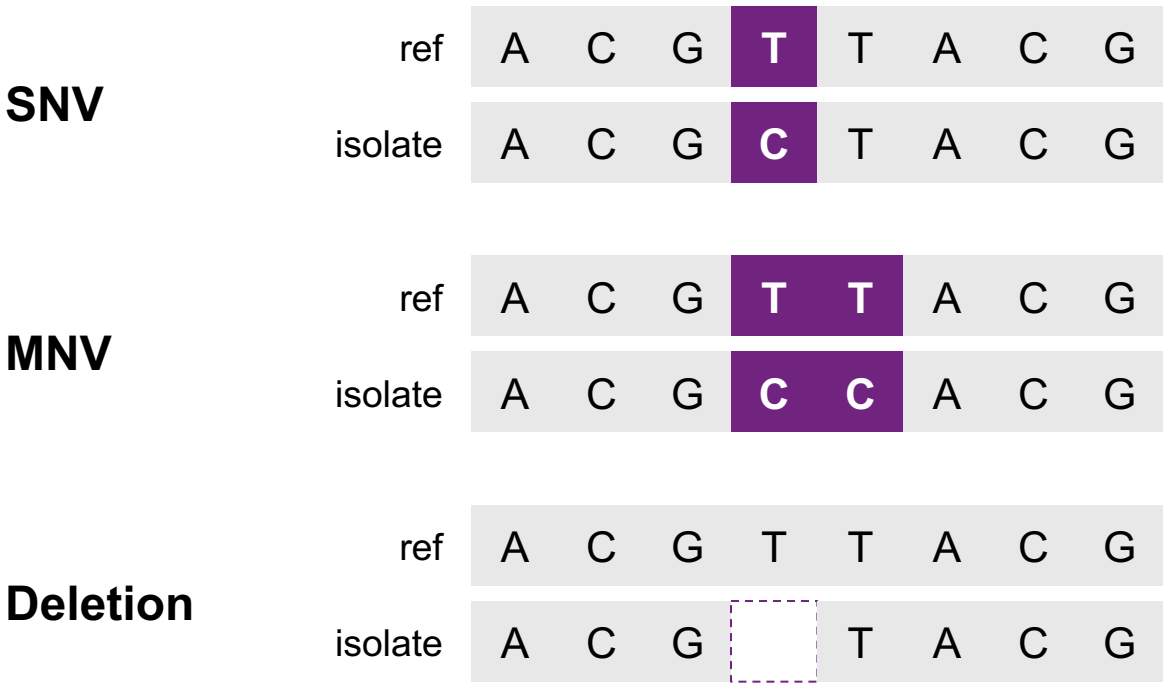
Where we are in the workflow



= this module (dashed outline = supplied or reference)

The case: where does the wolf differ from the corgi, base by base?

A variant is a difference from the reference



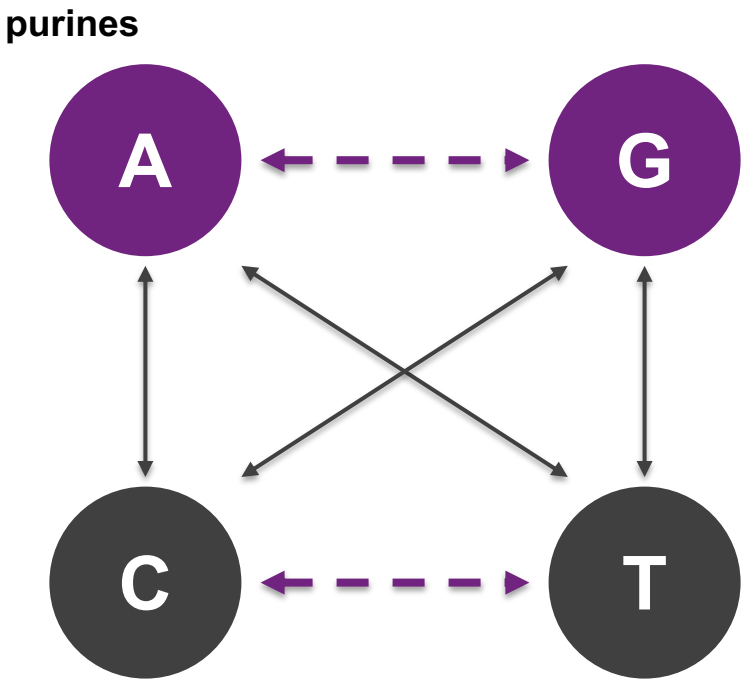
- Variant: any difference from the reference
- SNV: one base replaced by another
- MNV: several neighboring bases changed
- Indel: a small insertion or deletion (under 50 bp)
- 50 bp or more is a structural variant (SV): module 6

SNV or SNP? And how many differences?

- **SNV**: a single-base difference in one sample
- **SNP**: an SNV that is common in a population
- One isolate, no population: we say **SNV**
- How many differences between wolf and corgi? Lab 5

Type	Records
SNV	Lab 5
MNV	Lab 5
Indel, complex	Lab 5
Total	Lab 5

Transitions and transversions



Purple dashed: transitions (within a family)
Dark lines: transversions (between families)

- 4 possible transitions, 8 possible transversions
- Chance alone: about 1 transition per 2 transversions
- IRL Ts > Tv → Ts/Tv: a quick sanity check

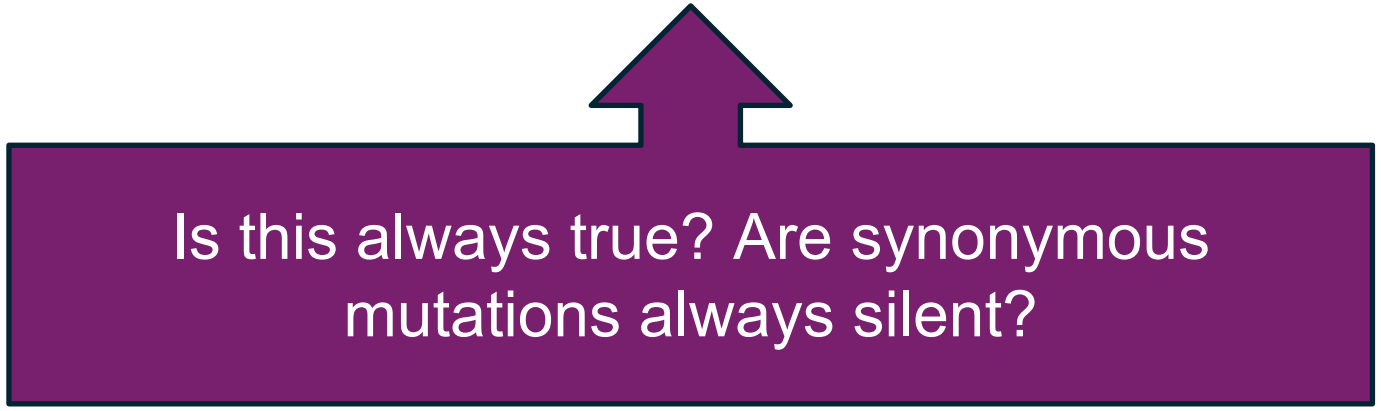
Effect of SNVs

Effect	Codon change	Protein	Consequence
Synonymous	GAA → GAG	Glu → Glu	No change in the protein
Missense	GAA → GAC	Glu → Asp	One amino acid differs
Stop gained	GAA → TAA	Glu → stop	Protein cut short
Frameshift	indel not a multiple of 3	all downstream changes	Usually breaks the gene

Outside genes (promoters, UTRs): the protein is unchanged, but expression may be alterd

Question:

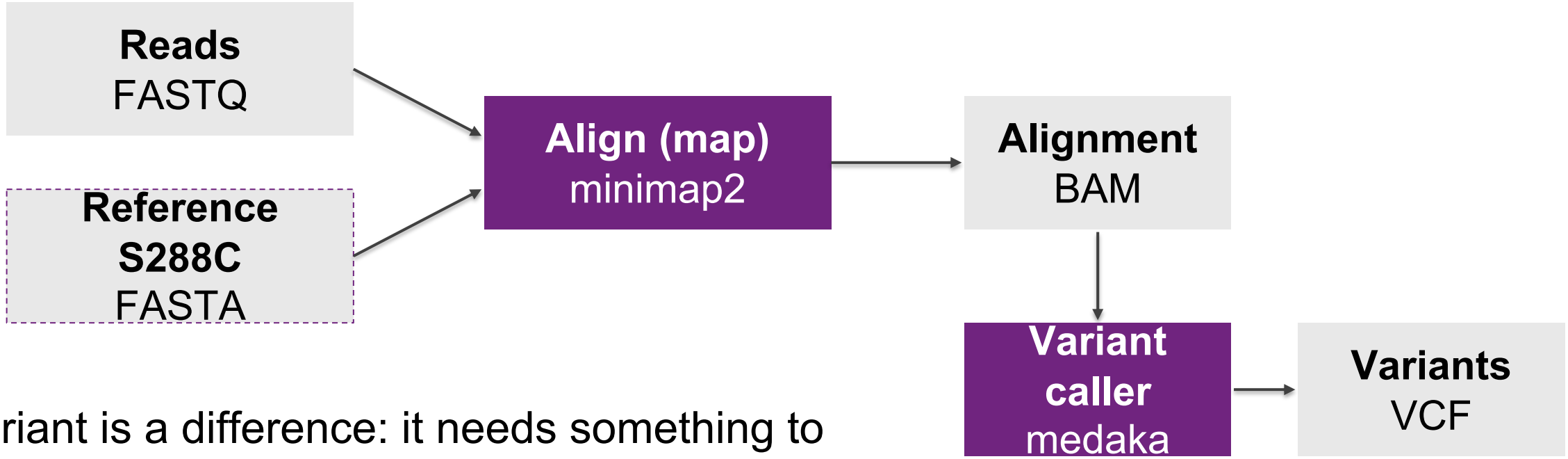
Effect	Codon change	Protein	Consequence
Synonymous	GAA → GAG	Glu → Glu	No change in the protein



What are variants used for?

- One gene: is it changed? (e.g. antifungal resistance via ERG11)
- A pathway: how many variants hit the genes of glycolysis?
- Strain fingerprinting: SNV profiles tell near-identical strains apart
- Evolution: variants are a molecular clock (is it a wine or a sake yeast?)
- Complex traits: GWAS link variants to traits such as growth rate

SNV calling needs a reference



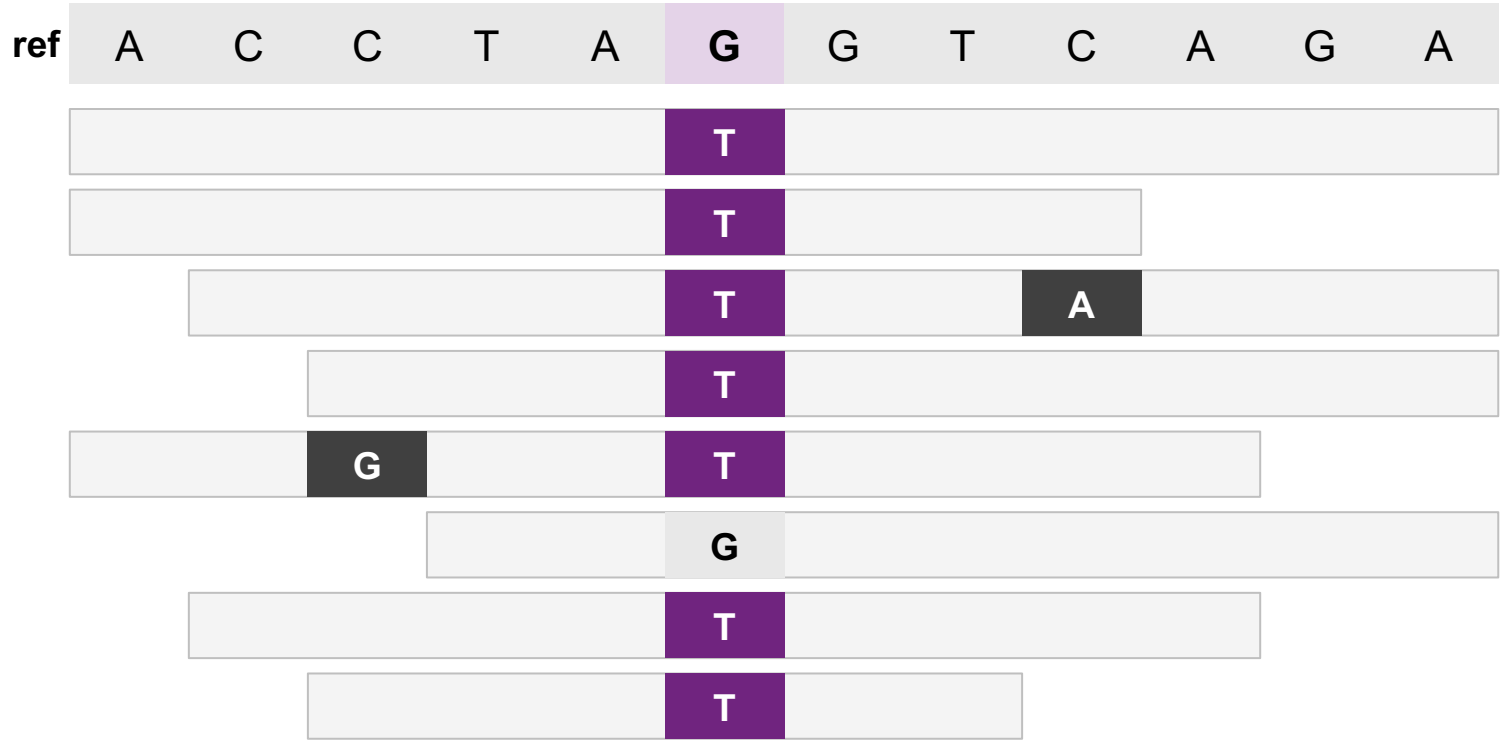
A variant is a difference: it needs something to differ from (here the corgi)

The reference also gives shared coordinates (example: IV:100,000)

Why use the reads, not the assembly?

Reads	Assembly
• Each read is an independent observation • Count the reads that support each base: depth and evidence at every site • Mixed evidence stays visible • Used for SNVs and small indels	• One consensus sequence: the vote is already taken • Per-read evidence is gone • A leftover assembly error looks like a real variant • Used for large rearrangements

The pileup: reading the evidence at one position



Purple: many reads agree on T, reference has G: a variant

Dark: one read only: likely sequencing error

Light purple: the reference base at the variant site

What makes a call reliable?

Depth
how many reads cover the site

Base quality
how sure is the sequencer about each base

Mapping accuracy
does the read belong here

Deep and agreeing (24 reads, 21 carry the called base)



Shallow and mixed (8 reads, 4 carry the called base)



Our R9.4.1 reads peak at about Q9 : some disagreement is expected

Variant callers

Caller	Reads	Notes
medaka_variant	ONT (long)	Neural network; fast; haploid caller; model should match chemistry. Used in the lab
Clair3	ONT or PacBio (long)	Deep learning; the long-read standard; recommended for diploids; slower to install and run

We use medaka because it is fast; Clair3 is the usual choice for diploid long-read data

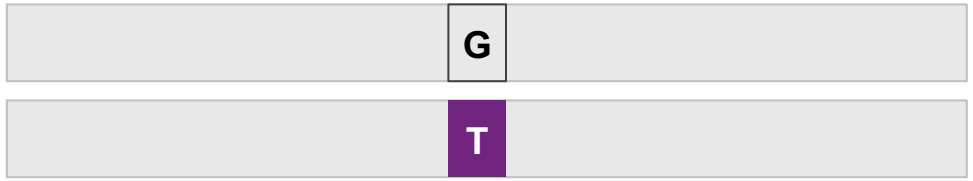
Ploidy decides the model

Haploid: one copy



One allele per site: the sample has T or it does not

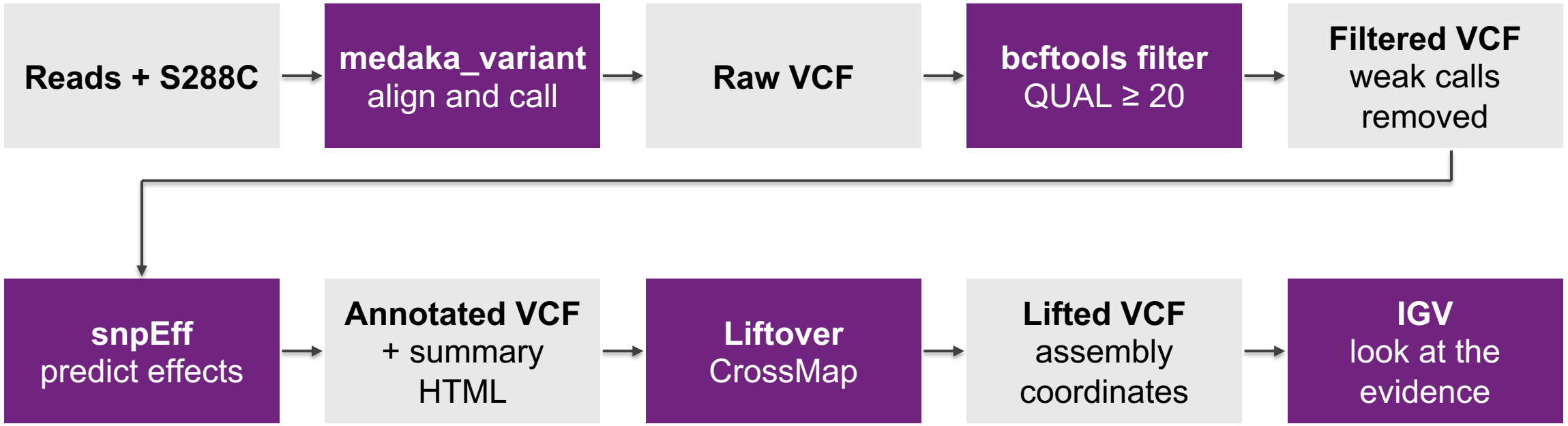
Diploid: two copies



Two alleles per site: heterozygous means the copies differ

- medaka_variant is a haploid caller; for diploids use Clair3
- BAL_1a: a homozygous diploid
- Is a haploid caller a fair model for a homozygous diploid?

From reads to variants: the whole lab in one picture



Purple: programs. Grey: files. Lab parts 1 to 5 follow this order

The alignment file: SAM and BAM

Read	Flag	Ref	Pos	MAPQ	CIGAR
r005	99	chr1	150	60	5M1I4M1D5M

- BAM = the binary, compressed form of SAM (one line per read)
- MAPQ: how sure the aligner is that the read belongs here
- CIGAR: the alignment as operations: M aligned, I insertion, D deletion, S soft-clipped
- Sorted by position and indexed (.bai) so tools and IGV can jump to a region
- Our BAM is the evidence: a real variant is seen in many reads

The VCF: only the differences

```
##fileformat=VCFv4.1
##medaka_version=2.2.2
##contig=<ID=IV,length=1531933>
#CHROM  POS  ID  REF  ALT  QUAL  FILTER  INFO  FORMAT  SAMPLE
IV  100000  .  C  T  30.0  PASS  .  GT:GQ  1:30
```

meta-information lines

column header

one line per variant

- VCF = Variant Call Format: plain text, tab-separated
- Matching sites are not listed: it holds only the differences

Reading one record

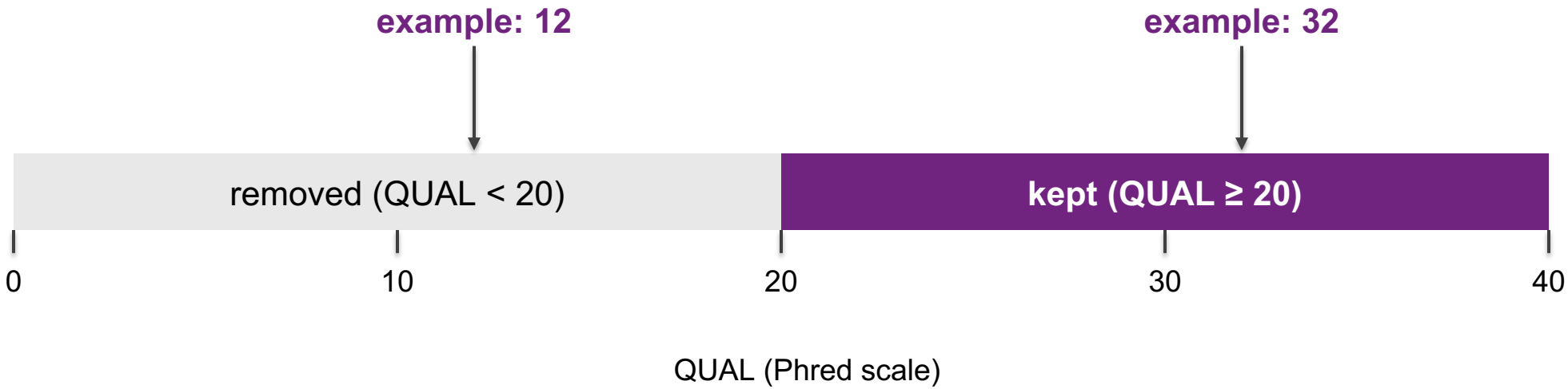
Column	Example value	Meaning
CHROM	IV	Chromosome
POS	100000	Position (1-based)
ID	.	Identifier (none)
REF	C	Base in the reference
ALT	T	Base in our isolate
QUAL	30.0	Phred-scaled confidence
FILTER	PASS	Passed the filters
INFO	.	Extra information (empty)
FORMAT	GT:GQ	What the sample column holds
SAMPLE	1:30	Genotype 1; genotype quality 30

Genotype (GT): haploid versus diploid

GT	Ploidy	Meaning
0	haploid	Reference allele
1	haploid	Variant (ALT) allele: what a haploid caller writes
0/0	diploid	Homozygous: both copies like the reference
0/1	diploid	Heterozygous: the two copies differ
1/1	diploid	Homozygous variant: both copies carry the ALT base

A haploid caller writes one number (0 or 1), never 0/1

QUAL: how confident is the caller?



QUAL 20 = 1 chance in 100 that the call is wrong; 30 = 1 in 1,000

Same Phred scale as read quality (Module 3)

We keep QUAL ≥ 20: how many calls survive? Lab 5

QUAL is the caller's own estimate, not a read count

Which variants can we trust least?

Reference: a run of six A



What different reads show



ONT (R9) reads misjudge homopolymer length (Module 3): indels are the least trustworthy calls

A false indel in a gene can look like a HIGH-impact frameshift

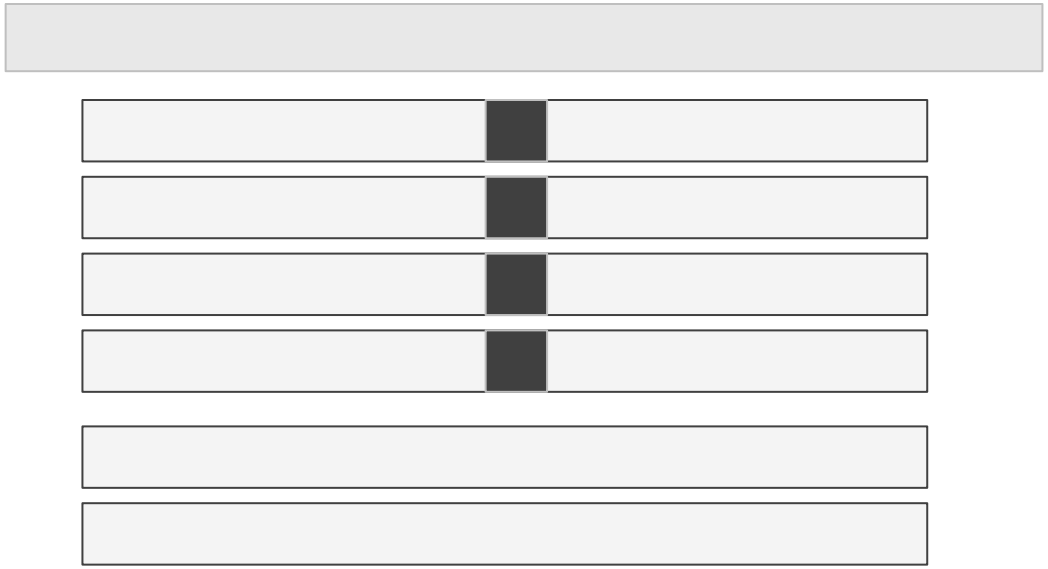
SNVs backed by many agreeing reads are the most reliable

Short reads need extra steps

Extra step	Problem it addresses
Duplicate marking	PCR copies count the same sequencing error more than once
Local realignment	Misaligned reads near indels create false SNVs
Base quality recalibration	The sequencer's quality scores are biased in systematic ways
Population structure and imputation	Shared haplotypes across samples improve uncertain calls

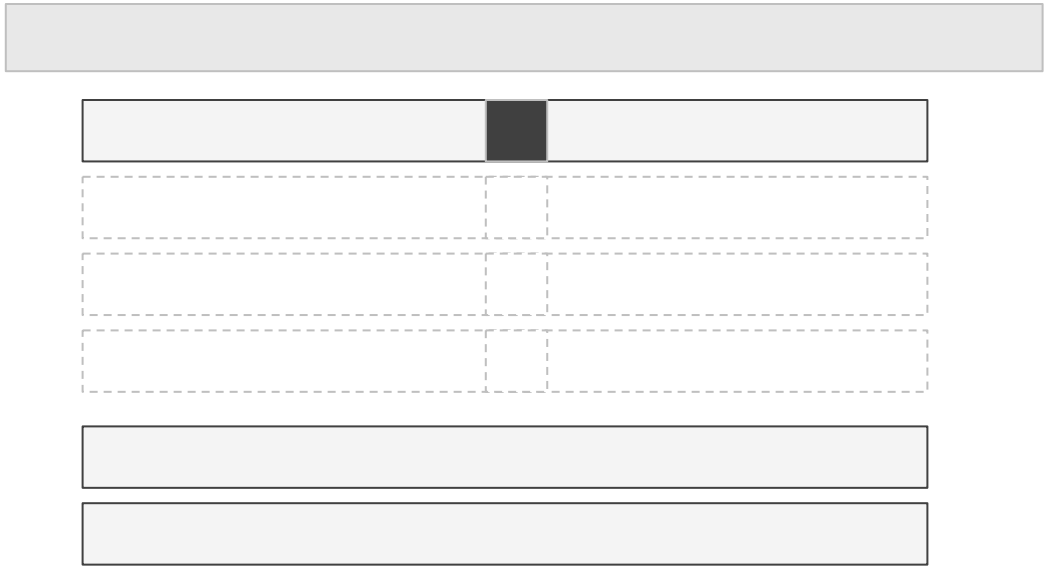
Duplicate marking

Before: copies counted as separate evidence



4 of 6 reads carry the base: looks like a variant

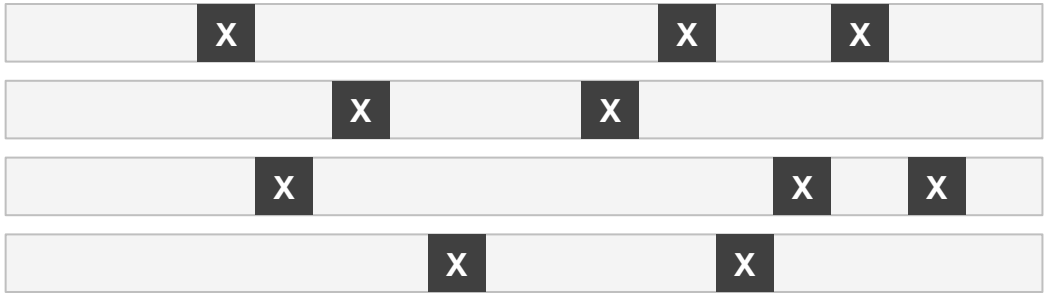
After: only one copy counted



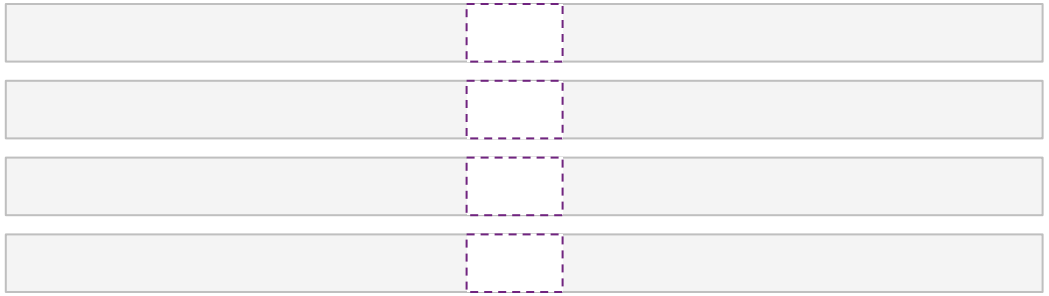
1 of 3 reads carries it: probably an error. Dashed = marked duplicate

Local realignment and base quality recalibration

Before: scattered mismatches



After: one agreed gap



Local realignment: realign reads around indels together, so false SNVs disappear (some callers now do this inside the call)

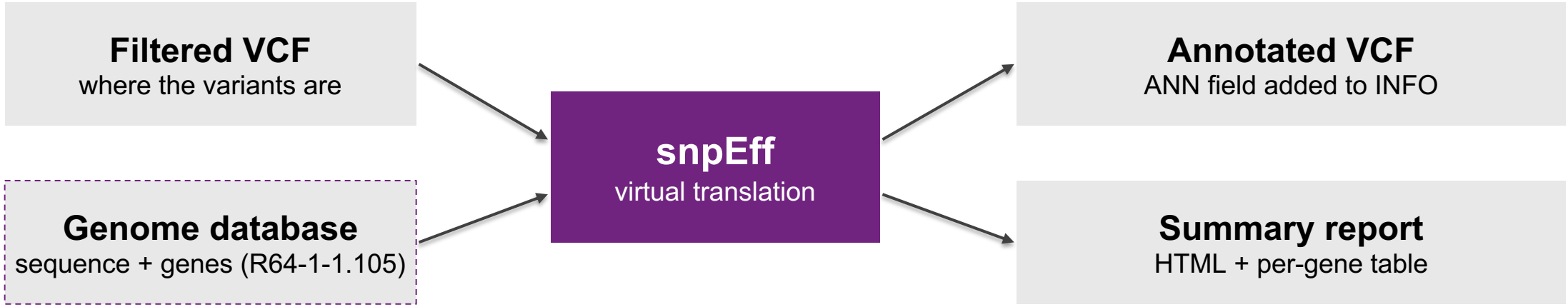
Base quality recalibration (BQSR): correct systematic biases in the sequencer's quality scores

Tools you may meet with short reads: variant calling

Tool	What it does
BWA-MEM2	Maps short reads to the reference
GATK HaplotypeCaller	Standard short-read caller (GATK Best Practices); clean-up steps: slides 24, 25, 26
DeepVariant	Deep-learning caller

Not covered in this workshop

snpEff: what does each variant do to a gene?



For each variant: which gene? which codon? which amino acid now?

A prediction from sequence alone, not a measurement

The database is the corgi's annotation (S288C), not the wolf's

Chromosome names in the VCF must match the database (I to XVI)

Impact classes: from gene breakers to volume knobs

HIGH	gene breakers: stop gained, frameshift
MODERATE	tweakers: missense (one amino acid changes)
LOW	silent: synonymous (same amino acid)
MODIFIER	volume knobs: promoter, UTR, between genes

Which effect dominates our HIGH calls, and can we trust it? Lab 5

Reading snpEff's ANN field

T	stop_gained	HIGH	GENEX	YDL999W	c.358C>T	p.Gln120*	120/310
allele	effect	impact	gene	gene ID	DNA change	protein change	amino acid position

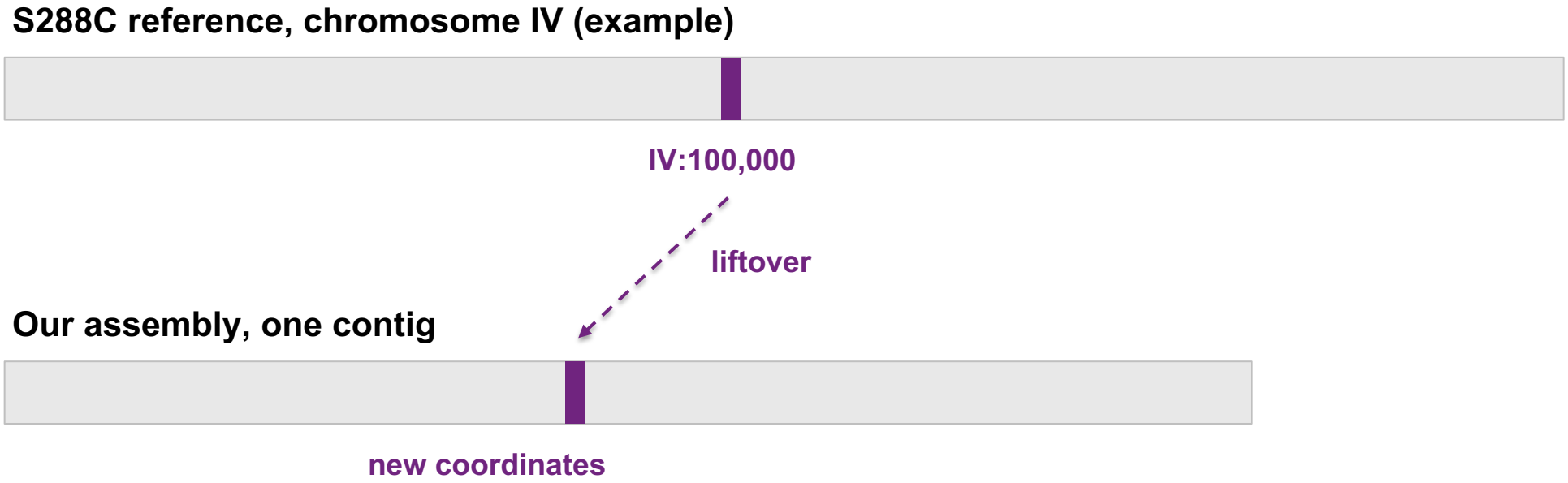
- Mock entry for an invented gene on an invented position. Fields are separated by "|"; one variant can have several entries (also neighboring genes, as MODIFIER)
- p.Gln120* means: amino acid 120 (glutamine) becomes a stop (*) in a protein of 310
- First entry: the most severe effect

Two fermentation genes to watch on chromosome XV

ADH1 (YOL086C)	GPD2 (YOL059W)
• Fermentative alcohol dehydrogenase • Acetaldehyde to ethanol • SGD S000005446 • The annotation example in Module 4 (S288C)	• NAD-dependent glycerol-3-phosphate dehydrogenase • gpd2Δ: impaired anaerobic growth • SGD S000005420

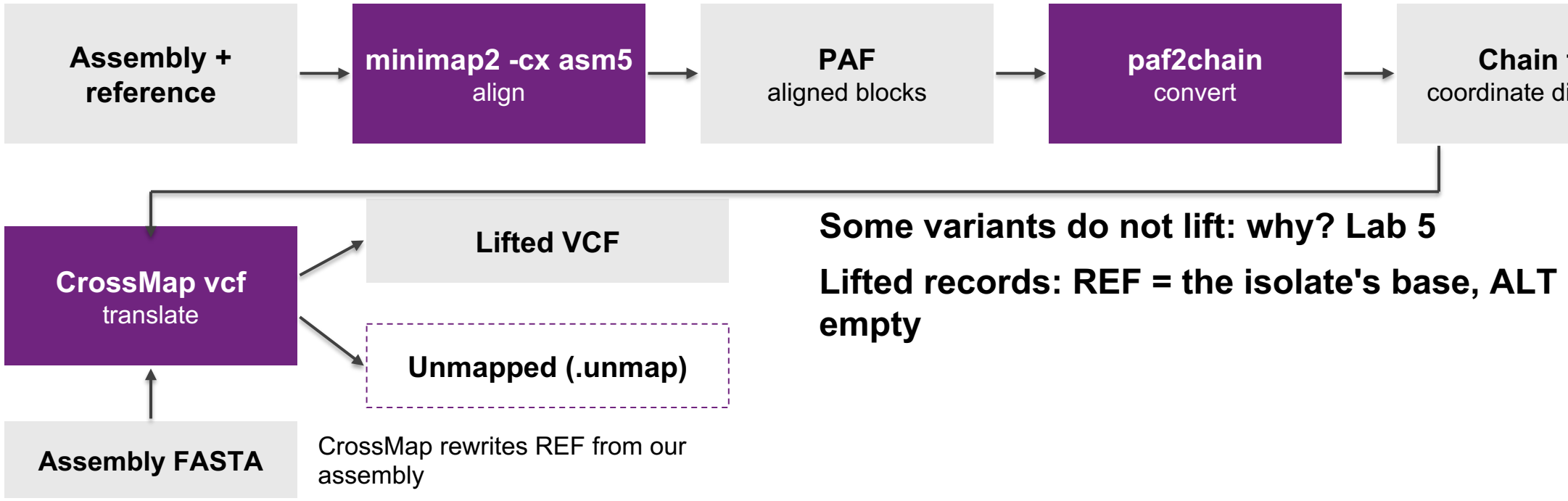
Does the wolf differ from the corgi here? Lab 5 Part 5.

Why lift variants over to the assembly?



- Variants are differences from S288C; our annotation is on our own contigs
- Liftover converts coordinates between assemblies

How liftover works: PAF, chain, CrossMap

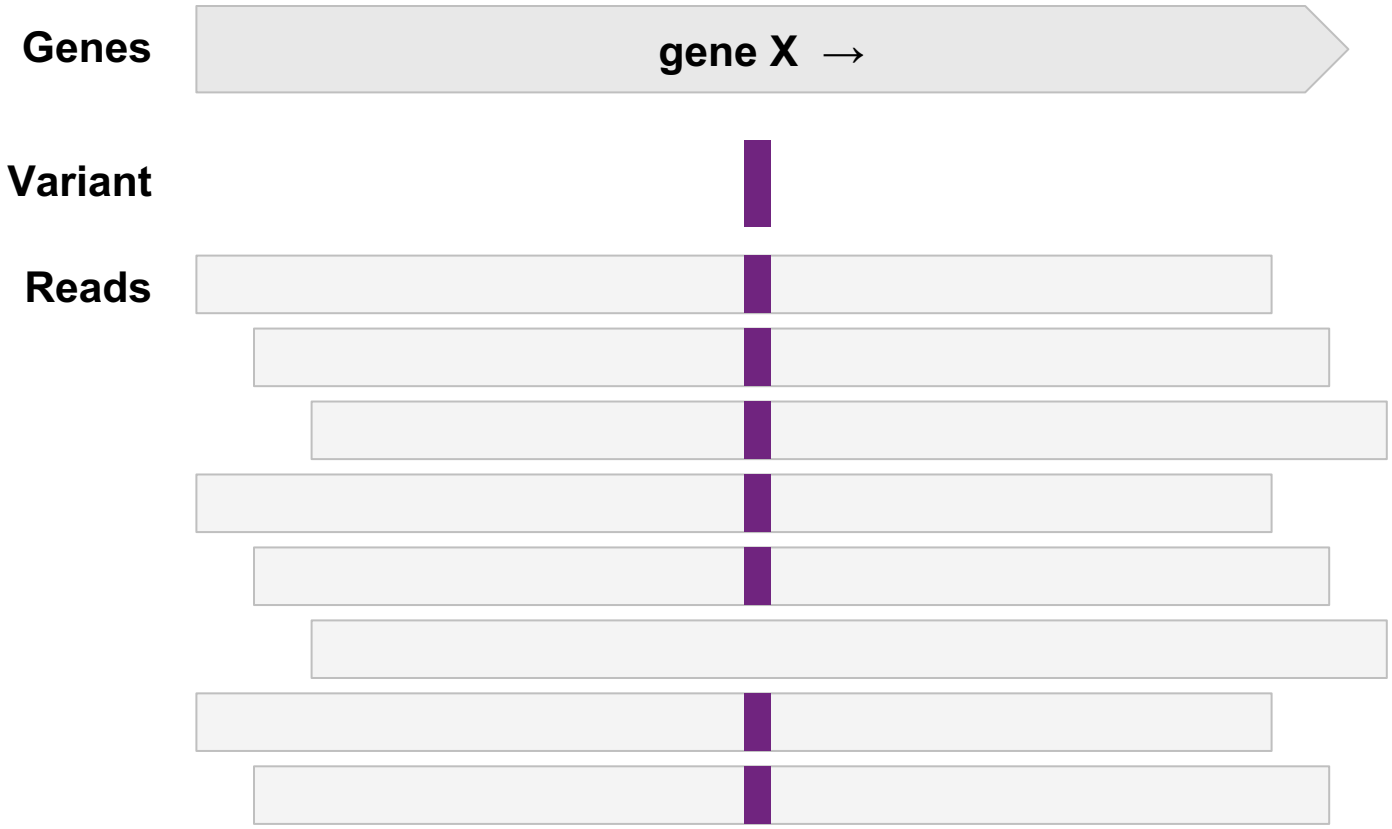


Some variants do not lift: why? Lab 5

Lifted records: REF = the isolate's base, ALT empty

PAF: minimap2's pairwise alignment format. **Chain file**: the coordinate dictionary CrossMap uses

Seeing is believing: variants in IGV



Purple: reads carrying the variant base

The evidence is the stack of reads, not the number in QUAL

Summary

- SNVs, MNVs and indels are differences from a reference; Lab 5 counts them in the wolf
- Call from reads aligned to a reference: medaka_variant is a **haploid** caller. Fair for BAL_1a? You decide
- A VCF lists only the differences: CHROM, POS, REF, ALT, QUAL, FILTER, INFO, FORMAT and GT
- Short reads need extra steps: duplicate marking, realignment, recalibration, imputation
- snpEff predicts impact (HIGH, MODERATE, LOW, MODIFIER): a prediction, not proof

Lab 5 preview: what you will do

1
Submit medaka_variant

2
Filter (QUAL ≥ 20) and read the statistics

3
snpEff, then copy the report with scp

4
Liftover onto the assembly

5
Open GPD2 and ADH1 in IGV

Questions are about reading results, not writing code

Coffee Break & Networking Session

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